



## THE HILLS GRAMMAR SCHOOL

### TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2004

# MATHEMATICS

**Time Allowed:**

Three hours (plus 5 minutes reading time)

**General Instructions:**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown for every question
- Begin each question in a new booklet

**Total marks – 120**

- Attempt Questions 1-10
- All questions are of equal value

|   | 01 | 02 | 02 | 03 | 03 | 04 | 04 | 05 | 06 | 06 |       |
|---|----|----|----|----|----|----|----|----|----|----|-------|
| Q | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | Total |
| M |    |    |    |    |    |    |    |    |    |    |       |

Students are advised that this is a Trial Examination only and cannot in any way guarantee the content or the format of the Higher School Certificate Examination.

**QUESTION 1 (12 marks)**

(a) Simplify  $3x - 6 - (4 - 2x)$ . **1**

(b) Fully simplify  $\sqrt{27} - \sqrt{3} + \sqrt{18}$ . **2**

(c) Solve  $|x - 1| = 2x - 1$ . **2**

(d) Express  $0.\dot{6}\dot{9}$  as a fraction in simplest terms. **2**

(e) Solve simultaneously

$$3x - y = 5$$

$$5x + 3y = -8$$

**3**

(f) Find the integers  $a$  and  $b$  such that

$$\frac{\sqrt{5}}{2 - \sqrt{5}} = a + b\sqrt{5}.$$

**2**

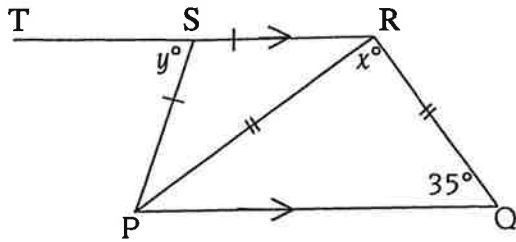
**QUESTION 2 (12 marks)**

If A (5, 3), B (-2, 5) and C (4, -3) are points on a number plane find the:

- |     |                                                                                       |   |
|-----|---------------------------------------------------------------------------------------|---|
| (a) | distance from A to C in surd form;                                                    | 2 |
| (b) | gradient of the line joining A and C;                                                 | 2 |
| (c) | equation of the line passing through A and C;                                         | 2 |
| (d) | perpendicular distance from B to the line passing through A and C;                    | 2 |
| (e) | area of the triangle ABC;                                                             | 2 |
| (f) | equation of the line perpendicular to the line AC passing through the midpoint of AC. | 2 |

**QUESTION 3 (12 marks)**

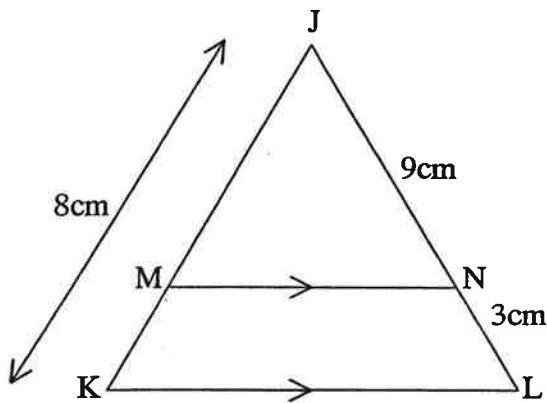
(a)



Find  $x^\circ$  and  $y^\circ$  in the diagram above.

**3**

(b)



In the triangle JKL,  $JK = 8\text{cm}$ ,  $JN = 9\text{cm}$  and  $NL = 3\text{cm}$ .

(i) Prove that  $\triangle JMN$  is similar to  $\triangle JKL$ .

**2**

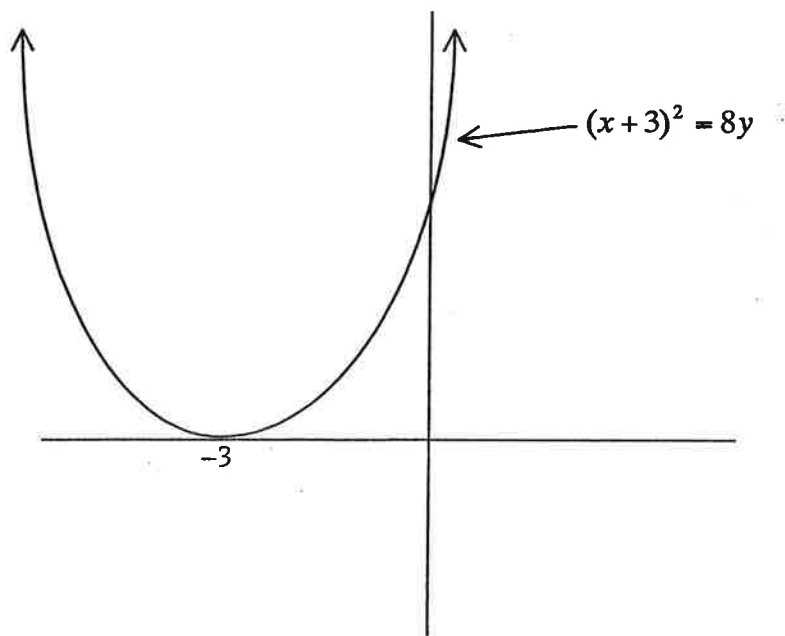
(ii) Find the length of MK.

**2**

**Question 3 continued**

(c) For this parabola find the:

- |       |                             |   |
|-------|-----------------------------|---|
| (i)   | co-ordinates of the vertex; | 1 |
| (ii)  | focal length;               | 2 |
| (iii) | co-ordinates of the focus;  | 1 |
| (iv)  | equation of the directrix.  | 1 |



**QUESTION 4 (12 marks)**

- (a) Find the exact arc length of a circle with radius 3cm and an angle subtended at the centre of  $60^\circ$ . **2**
- (b) Solve  $2\cos x = -\sqrt{3}$  for  $0 \leq x \leq 2\pi$ . **3**
- (c) (i) Sketch  $y = 3\cos 2x$  and  $y = x$  on the same set of axes for  $-\pi \leq x \leq \pi$ . **3**
- (ii) How many solutions does  $3\cos 2x = x$  have? **1**  
You are NOT required to find the solutions.
- (d) Find the equation of the normal to the curve  $y = x \sin x$  at the point  $(\pi, 0)$ . **3**

**QUESTION 5 (12 marks)**

(a) Solve  $3^{3x-2} = 11$  correct to 2 decimal places. 2

(b) If  $y = \log_e \left( \frac{2x+1}{3x-1} \right)$  find  $\frac{dy}{dx}$ . 2

Hint: use properties of logarithms before differentiating.

(d) Find (i)  $\int \frac{1}{x} + \frac{1}{x^2} dx$  1

(ii)  $\int \frac{4x}{x^2+1} dx$ . 2

(e) Differentiate with respect to  $x$

$y = x^2 \ln x$ . 2

(f) Show that  $\frac{2}{x+4} + \frac{3}{x-4} = \frac{5x+4}{x^2-16}$ . 3

Hence find  $\int \frac{5x+4}{x^2-16} dx$ .

**QUESTION 6 (12 marks)**

Evaluate each of the following integrals:

(a)  $\int x^6 - 2x^4 dx$ . 2

(b)  $\int_{-1}^2 \frac{x^4 + 3x^3 + x^2}{x} dx$ . 3

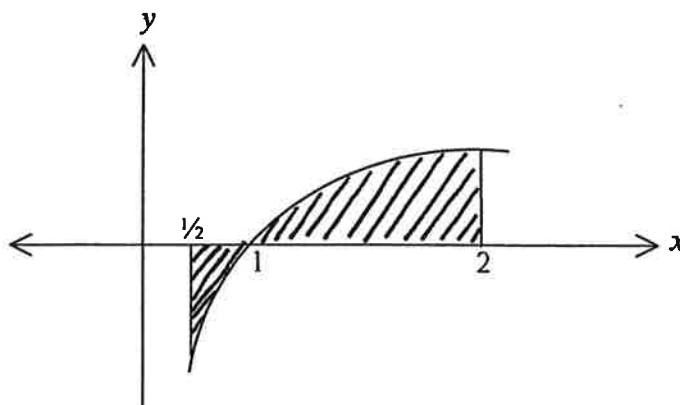
(c)  $\int \sqrt{3-x} dx$ . 2

(d) The table shows the values of a function  $f(x)$  for five  $x$  values. 3

|        |       |       |       |       |       |
|--------|-------|-------|-------|-------|-------|
| $x$    | 1     | 1.5   | 2     | 2.5   | 3     |
| $f(x)$ | 1.011 | 1.179 | 1.322 | 1.447 | 1.559 |

Approximate the value of  $\int_1^3 f(x) dx$  using the five function values and the Trapezoidal rule.

(e) Bob used  $A = \int_{\frac{1}{2}}^2 f(x) dx$  to find the shaded area in the diagram below. 2



Is this the correct method to use? If not, why not?

**QUESTION 7 (12 marks)**

(a) Consider the function  $f(x) = 4\sin 2x$ .

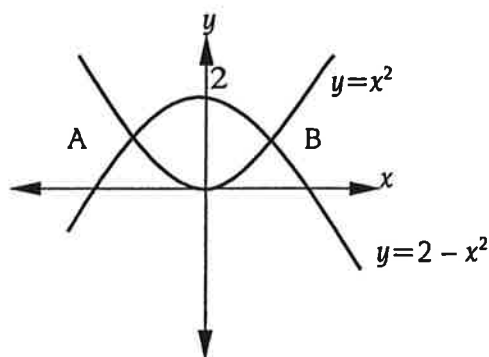
(i) Sketch  $f(x)$  for  $0 \leq x \leq 2\pi$ .

2

(ii) Using the result of part (i) and symmetry, or otherwise, find the area contained between the function  $f(x)$  and the  $x$ -axis for  $0 \leq x \leq 2\pi$ .

2

(b) The curves  $y = x^2$  and  $y = 2 - x^2$  intersect at two points,  $A$  and  $B$ .



(i) Find the coordinates of  $A$  and  $B$ .

2

(ii) Find the area bounded by the curves  $y = x^2$  and  $y = 2 - x^2$ .

3

(c) Find the volume formed when the curve  $y = \frac{\sqrt[3]{x}}{2}$  is rotated about the  $y$ -axis between  $y = 0$  and  $y = 2$ .

3

**QUESTION 8 (12 marks)**

- (a) An arithmetic series has a third term of 14 and a seventh term of 30.  
Find the first term and common difference of the series. **3**

- (b) Consider the series

$$-2 + 4(\pi - 3) - 8(\pi - 3)^2 + 16(\pi - 3)^3 - + \dots$$

- (i) Explain why the geometric series has a limiting sum. **1**

- (ii) Find the exact value of the limiting sum. **2**

- (c) The velocity, in metres per second, of a particle moving in a straight line is given by  
 $\dot{x} = t^3 - 7t - 7$ .

- (i) Find an expression for the displacement if the particle was initially 6 m to the left of the origin. **2**

- (ii) Calculate the acceleration of the particle after 3 seconds. **1**

- (iii) Find the displacement and velocity of the particle after 3 seconds. **2**

- (iv) Describe the motion of the particle when  $t = 3$  seconds. **1**

### **QUESTION 9 (12 marks)**

Observational evidence suggests that the level of “interest” students have in Mathematics lessons varies during a typical lesson according to  $I(t) = 5te^{-2t}$ , where  $I$  is the interest level, and  $t$  is time in hours.

- (a) Show that the first derivative,  $\frac{dI}{dt} = 5e^{-2t}(1 - 2t)$ . **1**
- (b) Show that the second derivative,  $\frac{d^2I}{dt^2} = 20e^{-2t}(t - 1)$ . **2**
- (c) At what time during the lesson is the student interest at a maximum? **3**
- (d) When is the level of interest in the lesson decreasing most rapidly,  
i.e. when is  $\frac{dI}{dt}$  at its minimum value? **3**
- (e) Find the location of any points of inflection on the curve  $I(t) = 5te^{-2t}$ . **1**
- (f) Using your findings from (a) – (e) above make a neat sketch of  $I(t)$  for  $0 < t < 2$ . Your sketch should include all important features. **2**

**QUESTION 10 (12 marks)**

- (a) According to one cosmological theory, there were equal amounts of uranium isotopes  $^{235}\text{U}$  and  $^{238}\text{U}$  at the creation of the universe in the “big bang”.

At present there are 137.7  $^{238}\text{U}$  atoms for each  $^{235}\text{U}$  atom. Using the half-lives

- 4.51 billion years for  $^{238}\text{U}$ ,
- 0.71 billion years for  $^{235}\text{U}$ ,

and assuming exponential decay, calculate the age of the universe.

**4**

- (b) Kippsy the mathematical kangaroo always hops (i.e. jumps) according to mathematical rules. One day, Kippsy decides to go hopping according to the following rules:

- The length of odd number hops (1<sup>st</sup>, 3<sup>rd</sup>, 5<sup>th</sup> hop etc.), in metres, is given by the arithmetic series  $t_n = 4 - (n - 1)$ , where  $n = 1, 3, 5, \dots$  is an odd number;
- The length of even number hops (2<sup>nd</sup>, 4<sup>th</sup>, 6<sup>th</sup> hop etc.), in metres, is given by the geometric series  $T_N = \frac{192}{63} \left( \frac{1}{2} \right)^{\frac{N-2}{2}}$ , where  $N = 2, 4, 6, \dots$  is an even number;
- If the length of a hop is negative according to the relevant series, Kippsy hops the prescribed distance *backwards*.

- (i) Write down the first term and common difference for the series  $t_n$ .

**1**

- (ii) Find the total distance hopped by Kippsy (in both directions) after the first 16 hops?

**3**

- (iii) Find where Kippsy is relative to her starting point after 12 hops.

**3**

- (iv) Describe the pattern of Kippsy’s hops after she has been hopping for a long time.

**1**

**End of paper**

Y-12 TRIAL 2 - UNIT  
2004

Question 1.

$$\begin{aligned} (a) \quad 3x - 6 - (4 - 2x) \\ = 3x - 6 - 4 + 2x \\ = 5x - 10 \end{aligned}$$

$$\begin{aligned} (b) \quad \sqrt{27} - \sqrt{3} + \sqrt{18} \\ = 3\sqrt{3} - \sqrt{3} + 3\sqrt{2} \\ = 2\sqrt{3} + 3\sqrt{2} \end{aligned}$$

$$\begin{aligned} (d) \quad n &= 0.69696969 \quad (1) \\ 100n &= 69.6969 \quad (2) \\ (2) - (1) \\ 99n &= 69 \\ n &= \frac{69}{99} \end{aligned}$$

$$\begin{aligned} (f) \quad \frac{\sqrt{5}}{2-\sqrt{5}} \times \frac{2+\sqrt{5}}{2+\sqrt{5}} \\ = \frac{2\sqrt{5} + 5}{4-5} \\ = \frac{2\sqrt{5} + 5}{-1} \\ = -2\sqrt{5} - 5 \end{aligned}$$

$$(c) \quad |x-1| = 2x-1$$

$$\begin{aligned} x-1 &= 2x-1 \\ 0 &= x \\ x &= 0 \end{aligned}$$

$$\begin{aligned} -x+1 &= 2x-1 \\ 2 &= 3x \\ x &= \frac{2}{3} \end{aligned}$$

$$\begin{aligned} (e) \quad 3x - y &= 5 \quad (1) \times 3 \\ 5x + 3y &= -8 \quad (2) \\ 9x - 3y &= 15 \quad (3) \\ (2) + (3) \\ 14x &= 7 \\ x &= \frac{1}{2} \\ \text{Subst } x = \frac{1}{2} \text{ into } (1) \\ 3(\frac{1}{2}) - y &= 5 \\ -y &= 3.5 \\ y &= -3.5 \end{aligned}$$

$$\boxed{a = -5 \quad b = -2}$$

Question 2.

$$A(x_1, y_1) \quad B(-2, 5) \quad C(x_2, y_2) \quad C(4, -3)$$

$$(a) \quad d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$(1) \quad d = \sqrt{(4 - 5)^2 + (-3 - 3)^2}$$

$$d = \sqrt{(-1)^2 + (-6)^2}$$

$$d = \sqrt{1 + 36} = \sqrt{37} \quad (1)$$

$$(b) \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-3 - 3}{4 - 5} \quad (1)$$

$$m = \frac{-6}{-1}$$

$$m = 6 \quad (1)$$

$$(c) \quad y - y_1 = m(x - x_1)$$

$$(1) \quad y - 3 = 6(x - 5)$$

$$y - 3 = 6x - 30$$

$$(1) \quad y = 6x - 27 \quad \text{or} \quad 6x - y - 27 = 0$$

$$\begin{aligned} (d) \quad \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right| &= \left| \frac{6(-2) + (-1)(5) + (-27)}{\sqrt{6^2 + (-1)^2}} \right| \quad (1) \\ &= \left| \frac{-44}{\sqrt{37}} \right| \approx 7.23 \text{ or } \frac{44}{\sqrt{37}} \quad (1) \end{aligned}$$

$$(e) \quad A = \frac{1}{2}bh$$

$$A = \frac{1}{2} \times \sqrt{37} \times \frac{44}{\sqrt{37}} \quad (1)$$

$$A = 22 \text{ u}^2 \quad (1)$$

$$(f) \quad \text{mid} = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left( \frac{5 + 4}{2}, \frac{3 - 3}{2} \right)$$

$$= \left( \frac{9}{2}, 0 \right) \quad (1)$$

$$= (4.5, 0)$$

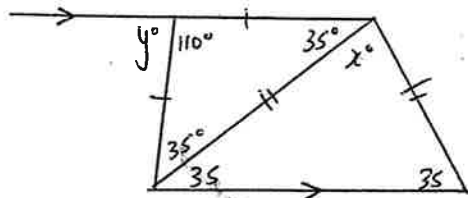
$$y - y_1 = m(x - x_1)$$

$$y - 0 = -\frac{1}{6}(x - 4.5)$$

$$y = -\frac{1}{6}x + \frac{3}{4} \quad (1)$$

### Question 3

(a)



$$x = 180 - 35 - 35$$

$$x = 110^\circ \quad (1)$$

$$y = 180 - 110$$

$$y = 70^\circ \quad (2)$$

(b) (i)  $\angle JMN = \angle JKL$  (corner  $\angle$  in  $\parallel$ ) A  
 $\angle JNM = \angle JLK$  (" " " " ) A

$\therefore \triangle JMN \parallel \triangle JKL$  (AAA) (1)

(ii)  $\frac{9}{12} = \frac{8-x}{8} \quad (1)$

$$72 = 12(8-x)$$

$$72 = 96 - 12x$$

$$12x = 24$$

$$x = 2 \quad (1)$$

(c) (i) vertex =  $(-3, 0)$  (1)  
 (ii) focal length = 2 units (1)  
 (iii) focus =  $(-3, 2)$  (2)  
 (iv) directrix  $y = -2$  (1)

### Question 4

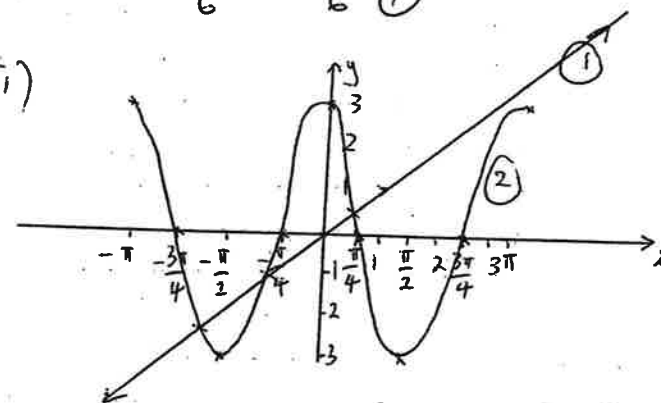
(a)  $l = r\theta$   
 $l = 3 \times \frac{60\pi}{180} \quad (1)$   
 $l = \pi \quad (1)$

(b)  $2 \cos x = -\sqrt{3}$   
 $\cos x = -\frac{\sqrt{3}}{2} \quad (1)$

$$x = 150^\circ, 210^\circ \quad (1)$$

$$x = \frac{5\pi}{6}, \frac{7\pi}{6} \quad (1)$$

(c) (i)



(ii) 3 solutions

(d)

$$y = x \sin x$$

$$u \quad v$$

$$u = x \quad v = \sin x$$

$$u' = 1 \quad v' = \cos x$$

$$y' = x \cos x + \sin x \quad (1)$$

$$y' = \pi \cos \pi + \sin \pi$$

$$y' = -\pi \quad y'_{\text{perp}} = \frac{1}{\pi} \quad (1)$$

$$y - 0 = \frac{1}{\pi}(x - \pi) \Rightarrow y = \frac{x}{\pi} - 1 \quad (1)$$

### Question 5

$$(a) \quad 3^{3x-2} = 11$$

$$\log_e 3^{3x-2} = \log_e 11$$

$$(3x-2)\log_e 3 = \log_e 11 \quad (1)$$

$$3x-2 = \frac{\log_e 11}{\log_e 3}$$

$$3x = \frac{\log_e 11}{\log_e 3} + 2$$

$$x = 1.39 \quad (1)$$

$$(b) \quad y = \log_e \left( \frac{2x+1}{3x-1} \right)$$

$$y = \log_e (2x+1) - \log_e (3x-1) \quad (1)$$

$$y' = \frac{2}{2x+1} - \frac{3}{3x-1} \quad (1)$$

$$(d) \quad (i) \quad \int \frac{1}{x} + \frac{1}{x^2} dx$$

$$= \log_e x - \frac{1}{x} + c \quad (1)$$

$$(ii) \quad \int \frac{4x}{x^2+1} dx$$

$$= 2 \int \frac{2x}{x^2+1} dx \quad (1)$$

$$= 2 \log_e (x^2+1) + c \quad (1)$$

$$(e) \quad y = x^2 \ln x$$

$$u = x^2$$

$$v = \ln x$$

$$u' = 2x$$

$$v' = \frac{1}{x}$$

$$y' = 2x \ln x + x^2 \cdot \frac{1}{x} \quad (1)$$

$$y' = 2x \ln x + x \quad (1)$$

$$(f) \quad \frac{2}{x+4} + \frac{3}{x-4}$$

$$= \frac{2(x-4) + 3(x+4)}{x^2-16} \quad (1)$$

$$= \frac{2x-8+3x+12}{x^2-16}$$

$$= \frac{5x+4}{x^2-16} \quad (1)$$

$$\therefore \int \frac{5x+4}{x^2-16} = \int \frac{2}{x+4} + \frac{3}{x-4} \quad (1)$$

$$= 2 \log_e (x+4) + 3 \log_e (x-4)$$

# Question 6

$$(a) \int x^6 - 2x^4 dx$$

$$= \frac{x^7}{7} - \frac{2x^5}{5} + C \quad (2)$$

$$(b) \int_{-1}^2 \frac{x^4 + 3x^3 + x^2}{x} dx$$

$$\frac{x^4}{x} + \frac{3x^3}{x} + \frac{x^2}{x} \quad (1)$$

$$\int_{-1}^2 x^3 + 3x^2 + x dx$$

$$= \left[ \frac{x^4}{4} + \frac{3x^3}{3} + \frac{x^2}{2} \right]_{-1}^2$$

$$= \left[ \frac{2^4}{4} + 2^3 + \frac{2^2}{2} \right] - \left[ \frac{(-1)^4}{4} + (-1)^3 + \frac{(-1)^2}{2} \right] \quad (1)$$

$$= 14 - \left[ \frac{1}{4} \right]$$

$$= 14 \frac{3}{4} \quad (1)$$

$$(c) \int \sqrt{3-x} dx$$

$$\int (3-x)^{1/2} dx \quad (1)$$

$$= \frac{(3-x)^{3/2}}{-1 \times 3/2} + C$$

$$= -\frac{2}{3} \sqrt{(3-x)^3} + C \quad (1)$$

$$(d) \text{Trape} = \frac{h}{2} \left[ (1.011 + 1.559) + 2(1.179 + 1.322 + 1.447) \right]$$

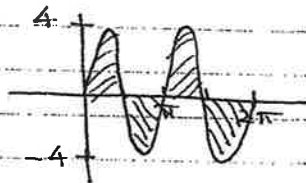
$$h = \frac{3-1}{4} \quad A = \frac{(1)}{4} [10.466] \quad (1)$$

$$h = \frac{1}{2} \quad A = 2.6165 u^2 \quad (1)$$

(e) (1) No, Area between  $\frac{1}{2}$  and 1 will be negative  $\therefore$  you need to integrate  $\int_{1/2}^1 f(x) dx + \int_1^2 f(x) dx$  (1)

Solutions & Marking Scheme

7 a) i)  $T = \frac{2\pi}{2} = \pi$  ①



ii)  $A = 2 \times 2 \times \int_0^{\pi/2} 4 \sin 2x \, dx$

$= 16 \int_0^{\pi/2} \sin 2x \, dx$  ①

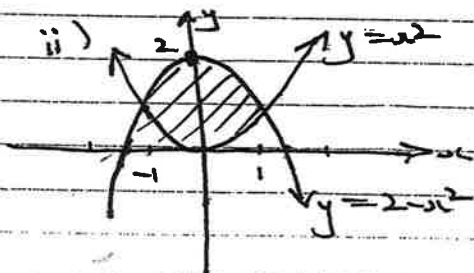
$= 16 \left[ -\frac{1}{2} \cos 2x \right]_0^{\pi/2}$

$= 16 \left[ -\frac{1}{2} \cos \pi + \frac{1}{2} \cos 0 \right]$

$= 16$  ①

b) i)  $y = x^2 = 2 - x^2 \Rightarrow 2x^2 = 2$   
 $x^2 = 1$

Points: (1, 1) & (-1, 1) ①



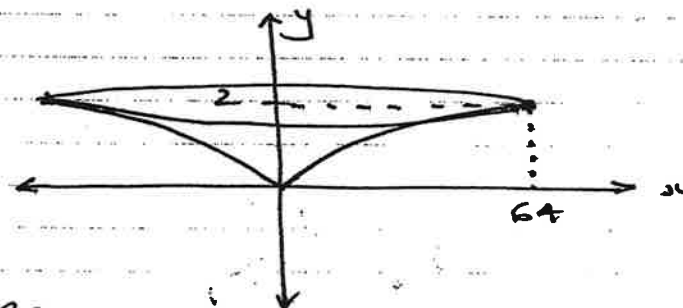
iii)  $A = 2 \int_0^1 [2 - x^2 - x^2] \, dx$

$= 2 \int_0^1 (2 - 2x^2) \, dx$   $\left\{ \begin{array}{l} a=0, \quad s=2 \\ b=1, \quad t=2 \end{array} \right.$  ①

$= 2 \left[ 2x - \frac{2x^3}{3} \right]_0^1$  ①

$= 2 \left[ 2 - \frac{2}{3} \right] = \frac{8}{3} \text{ units}^2$  ①

c)



~~Handwritten scribbles~~

$2y = x^{1/3} \Rightarrow 8y^3 = x$   
 $64y^6 = x^2$  ①

$V = \pi \int y^2 \, dx$  OR  $\pi \int x^2 \, dy$  ①

Here,  $V = \pi \int_0^2 x^2 \, dy$

$= \pi \int_0^2 64y^6 \, dy$  ①  $= \frac{8192\pi}{7}$  ①

$$= \pi \left[ \frac{64y^7}{7} \right]_0^2 = \frac{\pi \cdot 64 \cdot 128}{7}$$

$$= \frac{8192\pi}{7} \quad (1)$$

3

$$(8) a) T_n = a + (n-1)d$$

$$T_3 = a + 2d = 14 \quad (1)$$

$$T_7 = a + 6d = 30 \quad (2)$$

$$(2) - (1) \Rightarrow 4d = 16 \Rightarrow d = 4 \quad (1)$$

$$(1) \Rightarrow a + 8 = 14 \Rightarrow a = 6 \quad (1)$$

$$b) i) r = \frac{4(\pi-3)}{-2} = -\frac{8(\pi-3)^2}{4(\pi-3)} = \dots$$

$$r = -2(\pi-3) \approx -0.283$$

Since  $|r| < 1$ , G.P. has a limiting sum (1)

$$ii) S_{\infty} = \frac{a}{1-r} \quad (1)$$

$$= \frac{-2}{1-(-2(\pi-3))} = \frac{-2}{1+2(\pi-3)}$$

$$= \frac{-2}{2\pi-5} = S_{\infty}$$

$$\approx -1.56$$

~~$$S_{\infty} = \frac{a}{1-r} = \frac{6}{1-(-2(\pi-3))} = \frac{6}{1+2(\pi-3)} = \frac{6}{2\pi-5} \approx 1.56$$~~

$$(8) c) x = t^3 - 7t - 7$$

4

$$i) x = \int \left( \frac{dx}{dt} \right) dt$$

$$= \int (t^3 - 7t - 7) dt \quad (1)$$

$$= \frac{t^4}{4} - \frac{7t^2}{2} - 7t + c$$

$$x(0) = -6 \Rightarrow c = -6$$

$$x(t) = \frac{t^4}{4} - \frac{7t^2}{2} - 7t - 6 \quad (1)$$

$$ii) \dot{x} = \frac{d}{dt}(x) = 3t^2 - 7$$

$$\dot{x}(3) = 3 \cdot 3^2 - 7 = 20 \text{ ms}^{-2} \quad (1)$$

$$iii) x(3) = \frac{3^4}{4} - \frac{7 \cdot 3^2}{2} - 7 \cdot 3 - 6 = -\frac{153}{4} \text{ m} \quad (1)$$

$$= -38.25 \text{ m}$$

$$\dot{x}(3) = 3^3 - 21 - 7 = -1 \text{ ms}^{-1} \quad (1)$$

iv) Have  $x < 0$ ,  $v < 0$ ,  $a > 0$  so particle is to left of origin, moving left and accelerating, i.e. velocity becoming less negative. (1)

[5]

9 a)  $I = 5te^{-2t}$

$$\frac{dI}{dt} = 5t \cdot -2e^{-2t} + 5e^{-2t}$$

$$= 5e^{-2t}(-2t+1)$$

$$= 5e^{-2t}(1-2t) \quad (1)$$

b)  $\frac{d^2I}{dt^2} = 5e^{-2t} \cdot -2 + (1-2t) \cdot -10e^{-2t}$

$$= -10e^{-2t} - 10e^{-2t} + 20te^{-2t} \quad (1)$$

$$= 20te^{-2t} - 20e^{-2t} \quad (1)$$

$$= 20e^{-2t}(t-1)$$

c) Max. interest when  $I' = 0$ . (1)

$$\Rightarrow 5e^{-2t}(1-2t) = 0 \Rightarrow t = \frac{1}{2} \quad (1)$$

Concavity check.  $I''(\frac{1}{2}) = 20e^{-1}(-\frac{1}{2}) < 0$ . (1)

So, max. interest when  $t = \frac{1}{2}$ , i.e. 30 mins into lesson

d) Need minimum value for  $I'$ , i.e.  $I'' = 0$  (1)

$$I'' = 0 \Rightarrow t = 1. \quad (1)$$

Check for concavity.  $I''(1+E) > 0$  -ve / +ve  
 $I''(1-E) < 0$

So, min. value for  $I'$  when  $t = 1$  hr.

Interest decreasing most rapidly after 1 hr.

[6]

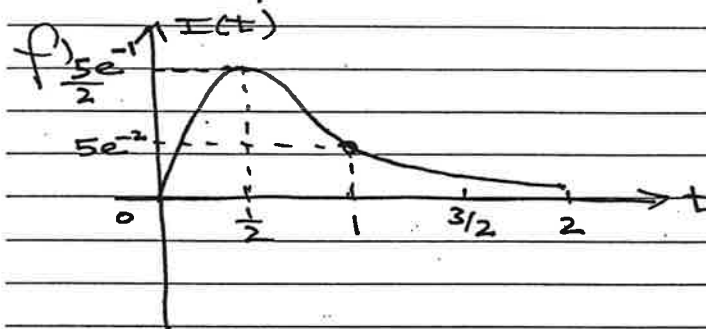
e) Point of inflection when  $I'' = 0$ , i.e.  $t = 1$ .

Check for concavity.

|       |       |     |       |
|-------|-------|-----|-------|
| $t$   | $1-E$ | $1$ | $1+E$ |
| $I''$ | $< 0$ | $0$ | $> 0$ |

(1)

So, point of inf. at  $t = 1$ .



(2)

10 a) For  $^{238}\text{U}$ , assume  $A_{238}(t) = A_0 e^{-kt}$

For  $^{235}\text{U}$ , assume  $A_{235}(t) = B_0 e^{-Lt}$ , where  $t$  is time in billions of years.

Know  $A_{238}(4.51) = A_0 = A_0 e^{-4.51k}$

$$\Rightarrow e^{-4.51k} = \frac{1}{2} \Rightarrow k = -\frac{1}{4.51} \ln\left(\frac{1}{2}\right)$$

Similarly,  $L = -\frac{1}{0.71} \ln\left(\frac{1}{2}\right)$   $k \approx 0.154$  (1)  
 $L \approx 0.976$  (1)

Now, have  $\frac{A_{238}(t)}{A_{235}(t)} = \frac{A_0}{B_0} e^{(-k+L)t} = 137.7$  (1)

But  $A_0 = B_0$ , so  $e^{(-k+L)t} = 137.7$

$$\Rightarrow t = -\frac{\ln(137.7)}{-k+L} \approx 5.99 \quad (1)$$

So, age of universe approx. 6 bill. yrs.



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(10) b) i)  $t_n = 4 - (n-1)$ ,  $n = 1, 3, 5, \dots$

[7]

Series is  $4 + 2 + 0 - 2 - 4 - \dots$

$a = 4, d = -2$ . (1)

ii) 16 hops  $\rightarrow$  8 from each series.

Odds:  $4 + 2 + 0 - 2 - 4 - 6 - 8 - 10 = -24$

Distance  $= 4 + 2 + 0 + 2 + 4 + 6 + 8 + 10 = 36\text{m}$  (1)

Evens:  $T_N = 192 \left(\frac{1}{2}\right)^{\frac{N-2}{2}}$ ,  $N = 2, 4, 6, \dots$

Series is  $\frac{192}{63} + \frac{192}{63} \left(\frac{1}{2}\right) + \frac{192}{63} \left(\frac{1}{2}\right)^2 + \dots$

$a = \frac{192}{63}$ ,  $r = \frac{1}{2}$ ,  $N = 8$ . (1)

$S_8 = \frac{192}{63} \left(1 - \frac{1}{2}^8\right) / \left(1 - \frac{1}{2}\right) \approx 6.07\text{m}$  (1)

Total distance hopped  $= 36 + 6.07 = 42.07\text{m}$ .

iii) Displacement after 12 hops  $= S_6 + S_6$   
(AP) (GP)

From (ii),  $S_6 = -6$  (1)

$S_6 = \frac{192}{63} \left(1 - \frac{1}{2}^6\right) / \left(1 - \frac{1}{2}\right) = 6$ . (1)

So, total displacement  $= -6 + 6 = 0$ ,  
i.e. Kippsey back at her starting point. (1)

iv) As  $N \rightarrow \infty$ ,  $T_N \rightarrow 0$  so  
Kippsey hopping backwards every 2nd  
hop, at ever increasing distances. (1)



The AGSM is a School of both The University of Sydney and The University of New South Wales

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